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Broadband, High Efficiency, Class J Power Amplifier Design Method with Compensating Drain-Source Capacitance

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A broadband, high efficiency, class J power amplifier (PA) design uses the relationship between drain efficiency and the ratio of drain-source capacitance to load impedance. In addition, multi-section matching is applied at the input and output to expand the bandwidth. A wideband class J PA designed with this approach and a Wolfspeed CGH40010F GaN HEMT achieved 40.0 to 41.5 dBm output power with 60 to 68 percent drain efficiency and an adjacent channel power ratio better than -30 dBc from 1.4 to 3.6 GHz.

With the rapid development of modern wireless communication technology, the amount of information transmitted is increasing and requiring wider bandwidth. This inevitably challenges the design of broadband PAs for wireless transmitters. The class J PA proposed by S.C. Cripps in 2006[1] uses drain-source capacitance, C_{ds}, to control harmonics at the output of a conventional class AB power amplifier to achieve high efficiency.[2] In recent years, several methods for expanding the bandwidth of class J amplifiers have been reported.[3-6] Saxena et al.3 designed a continuous class J PA using the nonlinear embedding approach, achieving between 63 and 72 percent drain efficiency from 1.3 to 2.4 GHz. Friesicke et al.[4] proposed a resistive-reactive class J PA with complex load impedances to explore efficiency-bandwidth tradeoffs in wideband class J designs with lossy second-harmonic loads. Andersson et al.[5] described a class J PA with dynamic load modulation to obtain output power and high efficiency over a certain dynamic range, and a broadband, high efficiency PA based on hybrid continuous modes with a phase shift parameter was reported by Huang et al.[6]

In this article, a design method based on traditional class J PA theory, which predicts high efficiency within a certain range of C_{ds}/R_L, is described. This work analyzes the trend of drain-source capacitance with the drain-source voltage and proposes a circuit to compensate transistor output capacitance so that C_{ds}/R_L can remain in the high efficiency range. A wideband J PA was designed and fabricated to validate the approach.

ANALYSIS AND DESIGN

Based on traditional theory, the transistor output voltage for an ideal class J PA can be expressed as[4]

$$\omega C_{ds} V_o(\theta) = I_{max}\left[1-\frac{\theta}{\pi}-\cos\theta\right]+ I_1\left[\cos(\theta+\phi)-\cos\theta\right] \text{ for } 0<\phi<\pi$$
$$= I_{max}\left[2-\frac{\theta}{\pi}\right]+ I_1\left[\cos(\theta+\phi)-\cos\theta\right] \text{ for } \pi<\phi<2\pi \quad (1)$$

where I_{max} is the maximum output current of the drain, I_1 is the output fundamental current, φ is the conduction angle, C_{ds} is the transistor drain-source capacitance and θ is a simple representation of the matching circuit phase, its value is determined by the parameters of each part of the output matching circuit.

The relationship between drain efficiency and X_{Cds}/R_L is shown in **Figure 1** under ideal conditions,[1] showing the drain efficiency remains at the ideal 78.5 percent when the value of X_{Cds}/R_L is between 1 and 2.5:

$$1 < 1/\omega C_{ds} R_L < 2.5 \quad (2)$$

This range provides additional space for the design of a broadband, high efficiency, class J PA.

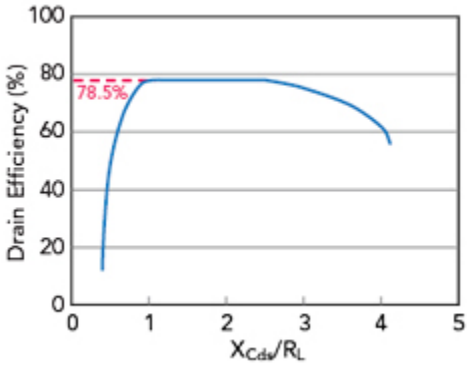


Figure 1 Drain efficiency vs. X_{Cds}/R_L.

Considering an ideal class B amplifier, R_L can be expressed as

$$R_L = R_{opt} = 2V_{dc} / I_{max} \quad (3)$$

where R_{opt} is the optimal load impedance and V_{dc} is the drain DC bias voltage. The transistor output capacitance in pF is

$$C_{out} = C_{outo} + A\left[1+\tanh(BV_{ds}+C)\right] \quad (4)$$

where C_{out} is the output capacitance of transistor and C_{outo}, A, B and C are constants associated with the transistor. For the Wolfspeed CGH40010F GaN HEMT, C_{outo} = 0.95, A = 1192.4, B = -0.0594714 and C = -2.94696.7

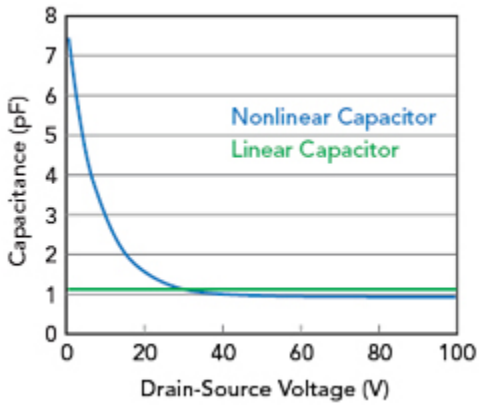


Figure 2 Output capacitance variation vs. drain-source voltage.

According to Equation 4, the relationship between the output capacitance and the drain voltage is shown in **Figure 2**. When the drain-source voltage V_{ds} is small, i.e., less than approximately 28 V, the transistor output capacitance decreases nonlinearly as the drain-source voltage increases. Beyond V_{ds} = 28 V, the capacitance reaches a constant value of 1.185 pF.

As the input signal power level has a direct impact on the magnitude of the drain signal power, changes in transistor output capacitance are discussed separately for low and high input power levels. For low input power, the effect of drain AC signal power can be ignored compared to the DC bias. The output capacitance

is approximately 1.185 pF for a DC bias of V_{ds} = 28 V. If the maximum current I_{max} is about 1.5 A, the theoretical high efficiency frequency range of operation is 1.44 to 3.61 GHz, found by applying Equations 2 and 3, and the corresponding drain efficiency is 78.5 percent. For large input power, the drain AC signal power has an obvious impact on the DC drain bias. For V_{ds} = 28 V, the output capacitance is nonlinear. The relationship between the capacitive reactance, X_{Cds}, and drain-source voltage is plotted in **Figure 3**, showing the capacitive reactance increases as frequency decreases and drain-source voltage increases.

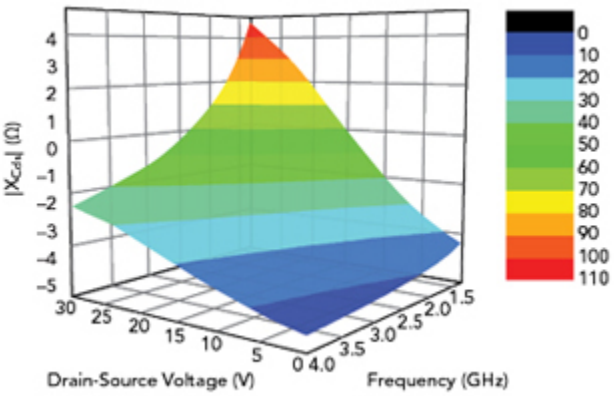


Figure 3 Output capacitive reactance variation vs. drain-source voltage and frequency.

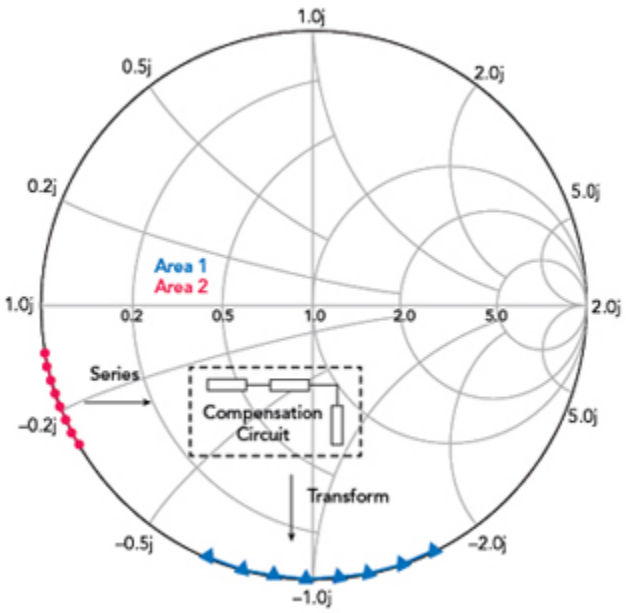


Figure 4 Impedance transformation of X_{Cds} to compensate for the nonlinear change in capacitance.

An external compensation circuit is needed to compensate for the nonlinear changes in capacitance so X_{Cds}/R_L can remain in the high efficiency region. The capacitive reactance, X_{Cds}, varies from -5.88 to -93.27 Ω. Area 1 in **Figure 4** shows that the impedance of the minimum capacitance value (1.18 pF) from 1.44 to 3.61 GHz varies from -93.27 to -37.20 Ω, which is ideal for high drain efficiency. Area 2 indicates that the impedance of the maximum capacitance value (9 pF) over the same frequency range varies from -14.73 to -5.88 Ω. In this work, the impedances of area 2 are transformed to those of area 1 (X_{Cds}/R_L = 1 to 2.5) by designing an appropriate output compensation circuit. The circuit in Figure 4 is a simple two-stage microstrip series section and shunt open stub. By selecting the appropriate microstrip parameters, this simple structure provides the desired compensation when connected at the output.

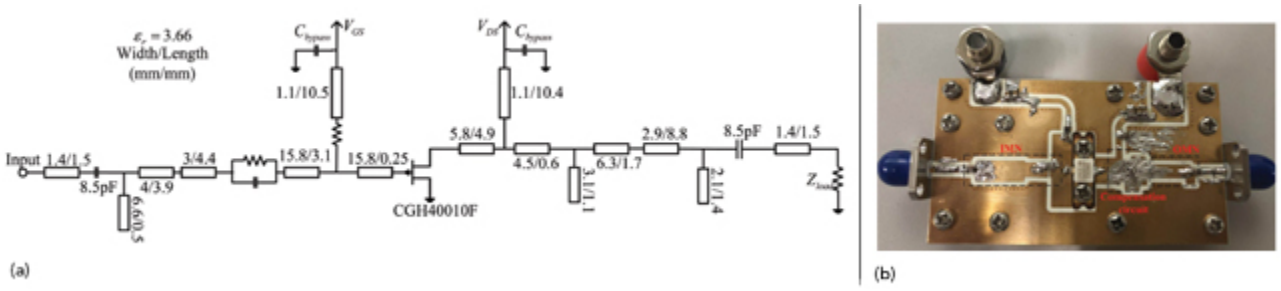


Figure 5 Class J PA schematic (a) and assembled amplifier (b).

MEASUREMENT RESULTS

To validate the design method, a broadband, high efficiency, class J PA was fabricated using the Wolfspeed CGH40010F GaN HEMT transistor and a Rogers 4350B substrate. The device was biased at V_{ds}= 28 V and V_{gs} = -2.8 V. To achieve broadband performance, the bias network was considered when performing input-output matching, which reduces the power leakage over a wide frequency band and improves output power and efficiency. The transistor packaging model was embedded in the output matching circuit to reduce the influence of parasitics. In addition, a multi-step matching method was used to increase bandwidth.

A schematic and a photograph of the class J PA are shown in **Figure 5**. **Figure 6** compares measurements with simulation for gain, output power and drain efficiency over the 1.4 to 3.6 GHz frequency band. Output power at 2 dB compression was 39.7 to 41.2 dBm, with drain efficiency of 60 to 68 percent and gain from 9.8 to 13.2 dB. **Figure 7** shows the ACPRs using a 5 MHz WCDMA signal, demonstrating -27.1 to -37.5 dBc over the band.

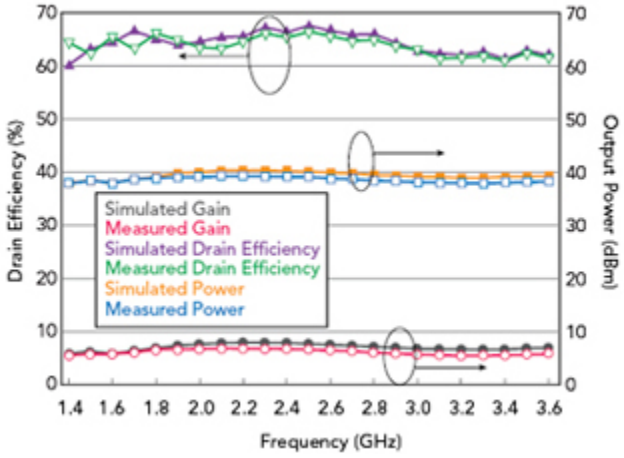


Figure 6 Measured vs. simulated drain efficiency, saturated output power and gain.

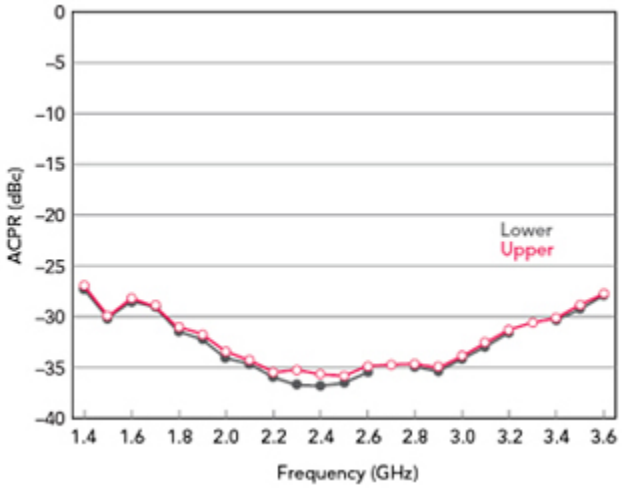


Figure 7 Measured ACPR.

Table 1 compares these results with the performance of other published broadband PAs. Although similar in most other respects, this design approach achieves greater bandwidth: approximately 88 percent compared to the 45 to 59 percent range reported for the other designs.

TABLE 1

COMPARISON WITH PRIOR WORK

Reference	Bandwidth (GHz)	Power (dBm)	Drain Efficiency (%)	Gain (dB)
3	1.3-2.4	40.0-41.3	63-72	10.5-11.2
8	1.9-3.0	40.0-42.2	58-72	11.5-12.2
9	1.6-2.8	40.0-42.5	67-81	11.9-15.2
10	2.0-3.5	40.0-41.3	64-76	11.8-13.8
This Work	1.4-3.6	39.7-41.2	60-68	9.8-13.2

CONCLUSION

An optimum output capacitor impedance space for a wideband, high efficiency, class J PA was defined and corresponding compensation circuit designed. A stepped impedance circuit was also adopted for input and output matching, accounting for transistor package parasitics to extend the bandwidth.

Acknowledgment

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